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How CEO Succession Influences Firm AI Innovation: The Role of Academic Experience Changes

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Dear editors,

We are pleased to submit our manuscript to *Journal of Business Research* for publication consideration. The manuscript is original and has not been submitted to any other journal.

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Despite growing interest in CEO succession and technological innovation, little is known about how changes in CEOs' academic experience shape emerging AI innovation. Drawing on the knowledge-based view (KBV), this study examines the relationship between CEO successions involving shifts in academic experience and post-succession AI innovation. Using a sample of 1,776 CEO succession events from Chinese high-tech firms, we find that transitions from non-academic to academic CEOs (NACA-to-ACA) significantly enhance AI innovation, whereas the reverse transitions (ACA-to-NACA) generally hinder it. Moreover, these effects are contingent on both internal and external knowledge conditions: the digital expertise–related power of the top management team and the strength of the regional AI knowledge ecosystem. Our findings contribute to the existing literature on CEO succession and AI innovation, providing valuable insights into emerging technological innovation in high-tech firms.

*This work was supported by Major Program of National Natural Science Foundation of China under Grant 72091311. The study's conclusions, however, are those of the authors, and do not necessarily represent those of the Foundation. The authors declare that there is no conflict of interest.

Thank you for the careful consideration of our work. We look forward to receiving comments from the editor and reviewers. Please contact me if you need any further information.

Best regards,

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How CEO Succession Influences Firm AI Innovation: The Role of Academic Experience Changes

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Declaration of Competing Interest

The authors declare that there is no conflict of interest.

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ABSTRACT

Despite growing interest in CEO succession and technological innovation, little is known about how changes in CEOs' academic experience shape emerging AI innovation. Drawing on the knowledge-based view (KBV), this study examines the relationship between CEO successions involving shifts in academic experience and post-succession AI innovation. Using a sample of 1,776 CEO succession events from Chinese high-tech firms, we find that transitions from non-academic to academic CEOs (NACA-to-ACA) significantly enhance AI innovation, whereas the reverse transitions (ACA-to-NACA) generally hinder it. Moreover, these effects are contingent on both internal and external knowledge conditions: the digital expertise–related power of the top management team and the strength of the regional AI knowledge ecosystem. Our findings contribute to the existing literature on CEO succession and AI innovation, providing valuable insights into emerging technological innovation in high-tech firms.

Keywords: CEO succession; academic experience; artificial intelligence innovation; strategic leadership; knowledge-based view

How CEO Succession Influences Firm AI Innovation: The Role of Academic Experience Changes

1. Introduction

Artificial intelligence (AI) has evolved from an emergent technology to a cornerstone of corporate strategy, with impactful AI innovation now a defining driver of the firm's competitive advantage and long-term performance (Krakowski et al., 2023; Zhai & Liu, 2023). AI innovation refers to the development of novel and improved artificial intelligence technologies or applications that enhance existing systems, introduce new capabilities, and solve complex problems (Babina et al., 2024; Liu et al., 2025). Unlike incremental technological improvements, AI innovation is characterized by high technical iteration, reliance on basic research, and extended development horizons, requiring top executives to make bold, long-term resource commitments amid market uncertainty (Lin & Maruping, 2025; Ångström et al., 2023). As global competition for AI innovation intensifies, understanding how top executives' attributes can shape firms' decisions to prioritize and scale AI innovation has become a critical research void for strategic management scholars.

CEOs are central strategic decision-makers and hold a pivotal position among top executives (Quigley & Hambrick, 2014; Zhu & Chen, 2016), so we primarily focus our research on CEOs. Academia plays a crucial role in driving the foundational technologies of AI, and AI innovation places distinctive demands on CEOs—such as deep technical literacy, tolerance for long-term research, and commitment to foundational breakthroughs—are often cultivated through academic careers (Chattopadhyay et al., 2025). Firms are increasingly appointing CEOs with academic experience in response to the growing strategic importance of AI innovation (Xu & Zhang, 2023). These academic CEOs introduce novel cognitive logics and advanced expertise into corporate

governance, reconfiguring firms' strategic priorities and innovation trajectories, thus encouraging investment in pioneering technological domains like AI (Liu et al., 2025; Ju et al., 2023).

Illustrative cases highlight this potential relationship: Fei-Fei Li, CEO of World Labs, is a tenured Professor of Computer Science at Stanford University and former Director of the Stanford Artificial Intelligence Laboratory (SAIL). Her longstanding research in computer vision and cognitive neuroscience equips her with deep insight into the core mechanisms underlying AI technologies and motivates the pursuit of frontier innovations in spatial intelligence. Founded only months earlier, World Labs rapidly scaled into a unicorn with a valuation exceeding US$1 billion. Similarly, Chen Tianshi, CEO of Cambricon and a former researcher at the Chinese Academy of Sciences has leveraged his academic expertise in AI chip architecture to drive significant technological breakthroughs, providing computational power for China's leading large-scale models. These technological advances propelled Cambricon's stock price to an all-time high above RMB 1,500 in 2025, with its total market capitalization reaching RMB 570 billion. Despite these prominent examples, prior research has rarely linked CEOs' academic experience to AI-specific innovation. We address this gap by investigating a critical yet overlooked question: How and when does CEO succession involving changes in academic experience affect a firm's AI innovation outcomes?

To address the questions, we integrate CEO background into the knowledge-based view (KBV) to provide a humanized framework for firm innovation. While the KBV asserts that the firm's competitive advantage lies in its capacity to generate, integrate, and apply knowledge (Grant, 1996), traditional literature emphasizes collective routines and organizational knowledge systems. Recent extensions highlight the influence of top executives in shaping knowledge processes (Singh et al., 2021). We advance this literature by conceptualizing CEO academic experience as a micro-

foundational antecedent to firm-level knowledge orchestration, thereby extending KBV to encompass individual cognitive characteristics.

CEO succession represents a critical turning point for organizational knowledge systems, as different leadership can reshape a firm's strategic orientations and organizational culture (Berns & Klarner, 2017; Choi & Benner, 2025), thereby influencing innovation performance. We focus on two types of CEO successions involving shifts in academic experience: NACA-to-ACA (from a non-academic CEO to an academic CEO) and ACA-to-NACA (from an academic CEO to a non-academic CEO). The former is theorized to enhance AI innovation by reconstructing research-oriented cognitive frames, facilitating cross-domain knowledge recombination, and strengthening long-term exploratory investments (Chattopadhyay et al., 2025). In contrast, the latter tends to prioritize operational efficiency and short-term profitability, which may disrupt established innovation trajectories and consequently weaken firms' AI innovation potential (Sauermann & Stephan, 2013; Shermon & Moeen, 2022).

Moreover, these effects are not deterministic but contingent on organizational and environmental moderators. Internally, top management teams (TMTs) with digital expertise can complement the knowledge-intensive leadership of academic CEOs, thereby ensuring strategic coherence and effective implementation of AI strategies (Filatotchev et al., 2023; Pinski et al., 2024). In ACA-to-NACA transitions, such TMT capabilities can serve as act as buffers to sustain innovation momentum (Firk et al., 2022). Externally, regional AI knowledge ecosystems—including AI-related institutions, talents, and networks—condition firms' access to external knowledge flows and collaborative opportunities (Igna & Venturini, 2023). In NACA-to-ACA transitions, rich regional ecosystems can strengthen CEOs' AI innovation agendas through cross-organizational learning (De Noni et al., 2018). Conversely, in ACA-to-NACA transitions, these

ecosystems can mitigate potential declines in innovation by maintaining access to external sources of novel knowledge.

Drawing on data from Chinese high-tech listed firms between 2013 and 2023, we conduct an empirical analysis by integrating unique information on AI patents, CEO academic experience, and CEO succession events. Our study makes several theoretical contributions. First, we reconceptualize CEO's academic experience as a strategic knowledge activation mechanism rather than a demographic attribute, thereby extending the KBV. This integration underscores the pivotal role of CEOs as innovation catalysts (Singh et al., 2021) and demonstrates how individual characteristics facilitate the strategic management of organizational knowledge for AI innovation. By bridging micro-level leadership traits with firm-level knowledge processes, we address KBV's traditional underemphasis on individual agency (Lord & Shondrick, 2011).

Second, we reveal the impact of CEO academic experience on AI innovation, thereby advancing previous studies that primarily consider AI's enabling role in innovation processes (Kakatkar et al., 2020; Mariani & Nambisan, 2021). While existing studies have largely focused on environmental, policy, and technological determinants of firm-level AI innovation, executive characteristics have received limited attention (Zheng et al., 2025). We highlight the role of CEOs' academic experience in shaping knowledge-intensive innovation and, by integrating TMT and regional knowledge characteristics, elucidate how CEO-driven knowledge orchestration interacts with broader innovation systems to shape AI outcomes. In doing so, we advance research at the intersection of leadership attributes and AI innovation.

Third, our findings challenge the prevailing view that emphasizes the disruptive effects of CEO succession (Zhang & Qu, 2016; Zhu & Shen, 2016). We show that transitions involving academic CEOs—NACA-to-ACA—can serve as catalysts for strategic renewal. Our results

demonstrate the positive role of academic CEO successions in fostering radical innovation and technological transformation (Chattopadhyay et al., 2025), thereby extending the literature on the dynamic implications of changes in academic experience during CEO succession processes.

2. Theoretical Background and Hypotheses Development

2.1 Toward a Leadership-Informed Knowledge-Based View (KBV)

The KBV asserts that knowledge is a firm's most strategically significant resource, serving as a primary driver of innovation and sustained competitive advantage (Grant, 1996). However, the KBV traditionally conceptualizes the firm as a unitary actor, often overlooking the critical role of individual agency—particularly top executives—in shaping, activating, and directing organizational knowledge resources (Ko & Choi, 2023; Singh et al., 2021).

To address this limitation, recent scholarship calls for a more micro-founded perspective that foregrounds individual cognition, experience, and behavior in strategic theory (Singh et al., 2021). As apex decision-makers, CEOs play a pivotal role in determining how knowledge is acquired, recombined, and leveraged. Their educational background and professional experience form cognitive templates that shape strategic direction, innovation-related risk-taking, and R&D investment decisions (Hambrick & Mason, 1984; He et al., 2021). For instance, academically trained CEOs typically exhibit stronger engagement with scientific knowledge, greater tolerance for uncertainty, and higher proclivity to long-term investments in radical innovation (Kaiser et al., 2018). Such CEOs are more likely to establish formal knowledge systems and institutionalized routines that translate complex and tacit insights into codified innovation practices. Accordingly, we integrate CEO background into the KBV and conceptualize it as a key mechanism through which organizational-level knowledge orchestration is enacted. This perspective repositions the CEOs as knowledge architects, offering a more compelling explanation for why firms with similar

knowledge bases may nonetheless exhibit divergent innovation trajectories.

2.2 Knowledge Reorientation: NACA-to-ACA CEO Succession and AI Innovation

We propose that NACA-to-ACA CEO successions are positively associated with post-succession firm-level AI innovation. Academic CEOs, shaped by rigorous scholarly training and immersed in knowledge-intensive environments, possess a distinctive cognitive orientation marked by epistemic curiosity, scientific reasoning, and long-term thinking (Chattopadhyay et al., 2025). Such leaders typically prioritize knowledge advancement and exploratory innovation over short-term operational goals. Upon succession, academic CEOs tend to break existing path dependencies established by non-academic predecessors and redirect strategic attention toward long-horizon R&D investment and transformative technological domains(Shen et al., 2020; Ju et al., 2023). This transition entails the reallocation of organizational resources from incremental improvement efforts toward high-risk, high-reward AI initiatives, thereby strengthening the firm's commitment to innovation goals (Xu & Zhang, 2023).

Moreover, AI innovation is contingent on both the systematic accumulation of codified knowledge and the dynamic integration of new scientific insights (Gama & Magistretti, 2025). Academic CEOs actively reshape organizational knowledge management systems to enhance innovative capacity. They institutionalize knowledge-based routines and processes that facilitate cross-functional knowledge flows in response to emerging developments in AI (Chattopadhyay et al., 2025). By encouraging broad employee engagement in idea generation and supporting open innovation platforms, dedicated R&D infrastructures, and organizational learning loops, academic CEOs help translate conceptual insights into deployable AI solutions (Yao et al., 2025), thereby accelerating both the scope and pace of AI innovation (Kaiser et al., 2018). In summary, we offer:

> ***Hypothesis 1a****: NACA-to-ACA CEO successions are positively associated with post-succession firm AI innovation.*

2.3 Innovation Backslide: ACA-to-NACA CEO Succession and AI Innovation

Lacking a deep-rooted commitment to scientific inquiry and long-term technological advancement, non-academic CEOs tend to deviate from the knowledge-oriented strategies of their predecessors and place greater emphasis on short-term financial metrics and market-driven objectives (Sauermann & Stephan, 2013). This strategic realignment often leads to reduced risk tolerance for exploratory AI programs and a reallocation of resources away from long-horizon R&D projects toward operational optimization and product enhancement (Xu & Zhang, 2023). Consequently, AI initiatives initiated under academic leadership may suffer from insufficient sponsorship or premature termination, thereby undermining the continuity of AI innovation (Ramachandran, 2018).

Furthermore, non-academic CEOs, guided by cost-efficiency imperatives, may underutilize or dismantle knowledge-intensive infrastructures developed under academic predecessors—including knowledge codification, dissemination and creation routines—that are critical for AI innovation (Ju et al., 2023). The cultural shift associated with transitions from academic to non-academic CEOs can create deep misalignments between top management and AI R&D teams (Sauermann & Stephan, 2013). Communication breakdowns, diminished morale, and reduced internal legitimacy of innovation teams may ensue, undermining both the retention of key innovative talent and the realization of firms' long-term AI vision. In summary, we propose:

> ***Hypothesis 1b****: ACA-to-NACA CEO successions are negatively associated with post-succession firm AI innovation.*

2.4 Digitally Empowered Leadership: TMT Power with Digital expertise as a Moderator

Within firms, the TMT acts as both a knowledge-processing unit and a governance mechanism. TMT digital expertise reflects the cumulative technical knowledge, experiential

capital, and strategic competence that TMT members possess concerning digital technologies, particularly AI infrastructure, data analytics, and implementation processes (Filatotchev et al., 2023). When this domain-specific expertise is coupled with TMT power—the ability to shape strategic agendas and influence organizational decision-making—it becomes a potent force in guiding firm-level innovation trajectories (Yoo et al., 2025). TMT members who simultaneously hold digital expertise and organizational power can provide continuity and legitimacy to AI investment decisions, thereby amplifying or buffering the effects of leadership transitions on AI innovation (Firk et al., 2022).

In NACA-to-ACA CEO successions, TMT members with digital expertise play a supportive role by amplifying the innovative strengths of academic CEOs and bridging potential gaps in strategy implementation (Bendig et al., 2023; Li et al., 2021). When endowed with organizational power, these TMT members can prioritize the alignment of R&D resources with innovation goals and ensure the effective deployment of AI initiatives (Pinski et al., 2024; Kraft et al., 2024). Their digital acumen further enables them to function as boundary-spanning agents who connect technical units with top leadership, fostering cross-functional learning and enhancing absorptive capacity to promote AI innovation (Yoo et al., 2025). During leadership transitions, their advocacy for knowledge-intensive innovation can generate positive spillover effects that enhance organizational engagement in AI initiatives, reinforcing the institutionalization of knowledge-based innovation routines (Filatotchev et al., 2023). We thus propose:

> ***Hypothesis 2a****: TMT power with digital expertise strengthens the positive association between NACA-to-ACA CEO succession and post-succession firm AI innovation.*

In contrast, during ACA-to-NACA CEO successions, TMT digital expertise plays a compensatory role that mitigates the potential erosion of AI innovation agendas. Digitally proficient and powerful TMT members can safeguard ongoing AI initiatives by emphasizing their

strategic importance and championing continued investments, reducing the risk of strategic drift when a non-academic CEO takes over (Chattopadhyay et al., 2025). They also sustain the internal knowledge systems and tacit routines necessary for AI development, even under short-term profit pressures, preventing disruptions to established innovation trajectories. Moreover, these members can minimize friction between innovation units and executive leadership, reduce organizational risk aversion, and sustain the long-term viability of AI innovation (Zhang et al., 2021). We thus offer:

> ***Hypothesis 2b****: TMT power with digital expertise weakens the negative association between ACA-to-NACA CEO succession and post-succession firm AI innovation.*

2.5 Embedded Advantage: The Moderating Role of Regional AI Knowledge Ecosystem

The regional AI knowledge ecosystem, defined as the geographically bounded concentration of AI-related institutions, talent, and infrastructures, facilitates dynamic inter-firm knowledge absorption, collaborative R&D, and benchmark-oriented learning (De Noni et al., 2018). The firm's embeddedness in such ecosystems affects its AI innovation capacity and the extent to which innovation agendas can be implemented during leadership transitions. Rich and well-developed ecosystems can enhance the innovative influence of academic CEOs; conversely, they can also stabilize AI innovation trajectories under the leadership of non-academic CEOs (Igna & Venturini, 2023).

In NACA-to-ACA CEO successions, the regional AI ecosystem serves as an accelerator by enabling rapid knowledge updating and minimizing learning cycles associated with AI innovation (De Noni et al., 2018). By fostering knowledge spillovers, this ecosystem enables incoming academic CEOs to integrate diverse perspectives and novel insights from peer firms into their AI innovation agendas (Doloreux & Mattson, 2008). Such interactions reinforce knowledge-based routines within the firm and enhance its absorptive capacity. Academic CEOs can further leverage

leading firms as benchmarks, drawing on best practices to advance superior AI solutions (Xu & Zhang, 2023). Moreover, the complementarity of ecosystem partnerships—whether through co-development agreements or research alliances—magnifies AI innovation outcomes by pooling specialized capabilities and sharing innovation risks (Zhang et al., 2019). For these reasons, we expect:

> ***Hypothesis 3a****: The regional AI knowledge ecosystem strengthens the positive association between NACA-to-ACA CEO succession and post-succession firm AI innovation.*

In contrast, during ACA-to-NACA CEO successions, the regional AI ecosystem provides a stabilizing counterbalance. Immersed in an environment that valorizes AI development, non-academic CEOs may become more receptive to innovation-driven imperatives (Hudson & Morgan, 2024). Through inter-firm interactions and regional collaboration networks, they can remain connected to frontier innovations and make more informed technology investment decisions (Igna & Venturini, 2023). Even in the absence of academic leadership, the complementary resource endowments provided by the ecosystem enable firms to maintain momentum in AI innovation through continuous access to technological knowledge, research infrastructure, and skilled talent (De Noni et al., 2018; Hagedoorn et al., 2018). For these reasons, we propose:

> ***Hypothesis 3b****: The regional AI knowledge ecosystem weakens the negative association between ACA-to-NACA CEO succession and post-succession firm AI innovation.*

3. Method

3.1 Sample

We test our hypotheses using a sample of high-tech A-share firms listed on the Shanghai and Shenzhen Stock Exchanges in China from 2013 to 2023. High-tech firms are identified based on national-level certifications from the National Enterprise Credit Information Publicity System (Li et al., 2024). These firms operate in environments characterized by rapid innovation cycles,

complex technologies, and intense competition (Pan et al., 2024). AI serves as a core general-purpose technology in high-tech firms, making them a suitable setting to examine how CEO succession affects AI innovation (Han et al., 2024). Our sample period aligns with a decade of national emphasis on AI following the launch of the "Internet + AI" initiative, during which AI patent applications increased by 368% compared with the prior decade.

CEO succession data were obtained from the China Stock Market and Accounting Research (CSMAR) database. We excluded interim CEO appointments, repeated successions by the same predecessor, incomplete cases, and events involving irregular compensation arrangements. CEO academic experience information was manually coded from executive resumes, annual reports, company websites, and supplementary online sources. AI innovation was measured using patent application data from the Chinese Research Data Services Platform (CNRDS), and firm-level financial and governance controls were drawn from CSMAR. To capture the longitudinal impact of leadership transitions on innovation performance, we tracked CEO succession events from 2013 to 2020 and collected AI innovation data spanning 2014 to 2023. The final dataset comprises 1,776 CEO succession events across 1,196 high-tech firms, of which 426 involved a change in CEO academic experience.

3.2 Variables and measures

3.2.1 Dependent Variable

Patents are widely regarded as a reliable indicator of innovation, and an growing number of studies adopt AI patent applications to evaluate technological advancement in the AI domain (Yang, 2022). Accordingly, we employed AI patent applications as a proxy for AI innovation, leveraging filing dates rather than grant dates to capture the timing of innovative breakthroughs and the contemporaneous influence of CEO succession (Bendig et al., 2023; Lou & Wu, 2021; Miric et al.,

2023). Owing to their structured documentation and institutional validation(Lou & Wu, 2021), AI patents were identified using the Key Digital Technology Patent Classification System issued by the China National Intellectual Property Administration (CNIPA) in 2023, which specifies AI-related International Patent Classification (IPC) codes.

We adopted a structured four-step procedure to construct firm-level AI innovation metrics. First, we compiled a comprehensive list of IPC codes related to AI technologies—such as G06F, G06K, and H02J—detailed in Table A1 (see Appendix A). Second, we matched these IPC codes with each firm's patent portfolio using an automated string-matching algorithm, supplemented by manual verification to ensure classification accuracy. Third, for patents with multiple IPC codes, we categorized them as AI-related if at least one code is included. Finally, we quantified post-succession AI innovation by tallying the total number of AI patent applications filed by each firm.

Consistent with existing literature suggesting that the impact of innovation activities typically materializes within a three-year horizon (De Noni et al., 2018), we measure *post-succession AI innovation* as the natural logarithm of the total number of AI patent applications in the three years following each CEO succession. For instance, a succession in 2013 corresponds to innovation output measured from 2014 to 2016.

3.2.2 Independent Variables

Following He et al. (2021) and Ju et al. (2023), we defined a CEO's academic experience as research-oriented professional experience prior to assuming the CEO role, including: (1) held a teaching or research position at a university; (2) worked at a public scientific research institution; or (3) engaged in research activities within a professional academic association. To examine the effects of changes in academic experience during CEO succession, we constructed two binary variables. *NACA-to-ACA CEO succession* is coded as 1 if the predecessor lacked academic

experience (Non-Academic, NACA) and the successor possessed one (Academic, ACA); otherwise, it is coded as 0. *ACA-to-NACA CEO succession* is coded as 1 if the predecessor had academic experience and the successor did not, and 0 otherwise.

Table A2 (see Appendix A) presents the distribution of predecessor-successor academic experience combinations over the study period. The majority of successions were NACA-to-NACA transitions (1,278 out of 1,776 events, or 71.96%). A smaller proportion were ACA-to-ACA transitions (72 events, or 4.05%). Notably, 426 succession events (23.99%) involved a change in academic experience, of which 305 were ACA-to-NACA transitions and 121 were NACA-to-ACA transitions. These distributions underscore the relative infrequency of academic experience shifts during CEO succession. Nevertheless, the presence of a meaningful subset of such transitions provides a valuable empirical foundation for investigating their potential influence.

3.2.3 Moderating Variables

To assess *TMT power with digital expertise*, we employed a two-step measurement process. First, we identified the digital expertise of individual TMT members based on three established criteria (Filatotchev et al., 2023; Firk et al., 2021): (a) an educational background in digital-related disciplines; (b) work experience in positions conducive to acquiring digital skills (e.g., Chief Information Officer); and (c) prior employment in digitally intensive industries (e.g., the IT sector). A TMT member was classified as having digital expertise if they satisfied at least one of these conditions. Tables A3-A5 (see Appendix A) provides a detailed listing of qualifying disciplines, job roles, and industries relevant to this classification.

In Chinese listed firms, the order in which top executives are listed does not consistently follow alphabetical conventions. Prior studies have validated the use of positional ranking within the TMT list as a reliable proxy for intra-team power dynamics (Ke et al., 2021). To quantify digital

expertise power, we first identified all TMT members with digital expertise, as described in the prior section. In cases where multiple such executives existed, we aggregated their individual power scores. Recognizing that variations in TMT size and the number of digitally skilled executives could introduce bias, we normalized the measure by calculating the proportion of total power held by executives with digital expertise. Each executive's power was computed based on their relative position in the TMT list, with higher-ranked positions assumed to confer greater influence. The TMT power score for each executive with digital expertise was then calculated using the following formula:

$$\text{TMT power} = 1 - (\text{TMT rank} - 1) / (\text{TMT size} - 1)$$

To measure the strength of the *regional AI knowledge ecosystem*, we quantified the number of AI-focused firms at the prefecture level across China. We adopted a systematic, keyword-based identification methodology aligned with prior research (Igna & Venturini, 2023; Li et al., 2021). Using the Tianyancha enterprise database (www.tianyancha.com), we analyzed the officially registered business scope descriptions, which provide detailed insights into firms' core activities and technological orientations. The identification process employed a textual analysis of business scope statements for all firms located within each prefecture-level city. A firm was classified as AI-focused if its registered business scope contained explicit references to AI-related technological indicators. These indicators encompassed both core AI technologies and applied domains, including semiconductor chips, image recognition systems, computer vision, speech recognition platforms, and sensor technologies. Additional AI-relevant keywords are listed in Table A6 (see Appendix A). We aggregated the total number of AI-focused firms identified in each prefecture. To account for skewed distributions and facilitate comparative analysis, we applied a natural logarithmic transformation to this count. The resulting metric reflects the relative density of AI-

oriented firms and serves as a proxy for the vitality of the regional AI knowledge ecosystem.

3.2.4 Control variables

Following prior research (Bendig et al., 2021; Lou & Wu, 2021; Zhang & Qu, 2016), we incorporated a comprehensive set of control variables spanning CEO characteristics, governance structures, and financial factors. At the CEO level, we controlled for *predecessor tenure*, measured as the number of years the predecessor served as CEO, given its influence on the successor's inherited legitimacy and strategic discretion (Zhang & Qu, 2016). *Successor duality* was coded as 1 if the successor simultaneously serves as board chair, reflecting consolidated decision-making authority and its implications for strategic consistency. *Successor gender* was coded as 1 for male to account for potential differences in leadership style and innovation propensity (Zhang et al., 2021).

At the governance level, we included *TMT size*, measured as the natural logarithm of the number of top management team members, to capture the balance between cognitive diversity and coordination efficiency in executive decision-making. *Board independence*, measured as the proportion of independent directors, reflects the quality of corporate oversight and its influence on strategic risk-taking (Bendig et al., 2023). *Ownership concentration*, operationalized by the equity share held by the largest shareholder, was included as it may shape managerial incentives and resource allocation toward innovation (Li et al., 2021).

At the firm level, we included prior-year ROA as a proxy for *pre-succession firm performance*, which can affect successor discretion and innovation strategy. *Firm size* was measured as the natural logarithm of total assets, and *R&D intensity* was calculated as as R&D expenditures divided by operating revenue to capture the firm's ongoing commitment to innovation. *Share of graduated employees*, defined as the proportion of employees holding tertiary education degrees, represents

the firm's human capital endowment critical for innovation (Rammer et al., 2022). *Market competition* was measured using the Herfindahl-Hirschman Index (HHI), calculated as the sum of squared market shares based on primary business revenue. Year and industry fixed effects were included to account for temporal and sectoral trends. All continuous variables were winsorized at the 1st and 99th percentiles to mitigate the impact of extreme values.

3.3 Models and estimation

Given the non-random nature of CEO succession, our analyses may be subject to selection bias and sample-induced endogeneity. To address these concerns, we employed the Heckman two-stage sample selection model (Zhang & Qu, 2016; Zhu & Shen, 2016). In the first stage, we aggregated all firm-year observations over the study period (n = 9,487) and employed a Probit model to estimate the probability of CEO succession. Following existing research (Connelly et al., 2022), we used the *peer CEO succession rate*, measured as the proportion of other firms within the same two-digit industry experiencing CEO succession in a given year, as the instrumental variable. This variable is theoretically justified because firm actions are shaped by industry context, while industry-level averages are unlikely to directly influence a focal firm's innovation outcomes. As expected, peer CEO succession rate was significantly and positively associated with the probability of CEO succession ($\beta = 3.142$, $p < 0.01$). We then calculated the Inverse Mill's Ratio from the first-stage model and included it as a control variable in the baseline regression model.

To further address potential endogeneity related to the selection of successors with academic experience, we implemented a second Heckman selection model focused on the 1,776 firm-year observations involving CEO succession. We employed the *proportion of regional academic CEOs* as the instrumental variable, defining it as the average proportion of academic CEOs from other firms within the same province in the previous year (Ma et al., 2020). The regional prevalence of

academic CEOs reflects labor market characteristics and institutional norms related to executive selection, but it is unlikely to influence a focal firm's AI innovation outcomes. The instrument variable was significantly and positively associated with the likelihood of appointing an academic CEO ($\beta = 2.205$, $p < 0.01$). The Inverse Mill's ratio from this selection equation was also incorporated into the models on post-succession AI innovation. Detailed estimation results from both first-stage models are reported in Table A7 (see Appendix A).

Finally, we included year and industry fixed effects and estimated all models using robust standard errors clustered at the firm level to account for within-firm correlation and heteroscedasticity. Given the presence of interaction terms, multicollinearity may be a concern (Haans et al., 2016). We centered all variables before computing interactions and calculated variance inflation factors (VIFs). The maximum VIF was 2.44, well below the widely accepted cutoff of 10, suggesting that, multicollinearity is unlikely to bias our results.

4. Results

4.1 Baseline results

Table 1 reports the descriptive statistics and pairwise correlation coefficients for all variables included in the analysis. Table 2 presents the regression results from the Heckman selection models examining the impact of CEO succession on post-succession AI patent applications. Model 1 serves as the baseline specification and includes only control variables. The results from Model 1 indicate significant positive associations between AI innovation and several firm-level characteristics, including board independence, share of graduated employees, ownership concentration, firm size, pre-succession firm ROA, and R&D intensity. In contrast, market competition shows a significant negative relationship with AI innovation outcomes.

Models 2-4 sequentially introduce the hypothesized explanatory variables related to CEO

succession characteristics and organizational context. Model 5 presents the full specification, incorporating all control and focal variables to test the complete set of hypotheses. The results from Model 5 are broadly consistent with those of the preceding models, reinforcing the robustness and stability of our main findings. These results suggest that CEO succession characteristics, when situated within the broader firm and regional context, have systematic and meaningful implications for AI innovation outcomes in high-tech firms.

[Insert Table 1 and Table 2 about here.]

We find support for Hypothesis 1a, which examines the effect of NACA-to-ACA CEO succession on AI innovation. In Model 2, NACA-to-ACA CEO succession shows a positive and significant relationship with AI innovation ($\beta = 0.451$, $p < 0.01$). Similarly, Hypothesis 1b is supported, as the coefficient for ACA-to-NACA CEO succession in Model 2 is negative and significant ($\beta = -0.279$, $p < 0.01$), indicating a detrimental effect on AI innovation. In economic terms, compared to successions without changes in academic experience, NACA-to-ACA CEO successions are associated with a 56.99% increase in AI patent applications ($e^{0.451}$-1), whereas ACA-to-NACA transitions correspond to a 24.35% decrease in AI patent applications (1-$e^{-0.279}$).

Hypothesis 2a proposes that TMT power with digital expertise positively moderates the relationship between NACA-to-ACA CEO succession and AI innovation. In Model 3, the interaction term NACA-to-ACA CEO succession × TMT power with digital expertise is positive and significant ($\beta = 1.859$, $p < 0.01$). Figure 1 illustrates this interaction effect, showing that when TMT members with digital expertise hold greater power, the positive relationship between NACA-to-ACA CEO succession and AI innovation is amplified. Conversely, firms with low TMT power exhibit minimal variation in this relationship. These findings support Hypothesis 2a.

Extending the analysis to the inverse scenario, Figure 2 reveals that during ACA-to-NACA

CEO succession events, firms with low TMT power with digital expertise background a notably steeper decline in AI innovation. In Model 3, the interaction term ACA-to-NACA CEO succession × TMT power with digital expertise is positive and significant ($\beta = 0.534$, $p < 0.01$). This indicates that TMT power with digital expertise mitigates the negative effect of ACA-to-NACA CEO succession on AI innovation, thereby supporting Hypothesis 2b.

[Insert Figure 1 and Figure 2 about here.]

Hypothesis 3a posits that a regional AI knowledge ecosystem strengthens the positive relationship between NACA-to-ACA CEO succession and AI innovation. In Model 4, the interaction term NACA-to-ACA CEO succession × regional AI knowledge ecosystem is positive and significant ($\beta = 0.111$, $p < 0.05$), indicating a reinforcing effect on AI innovation. This result aligns with Hypothesis 3a, as illustrated in Figure 3. Model 4 also shows a significant interaction term for ACA-to-NACA CEO succession × regional AI knowledge ecosystem ($\beta = 0.102$, $p < 0.01$). In Figure 4 regions with a robust AI knowledge ecosystem background a less pronounced decline in AI innovation following ACA-to-NACA CEO succession. These findings support Hypothesis 3b.

[Insert Figure 3 and Figure 4 about here.]

4.2 Robustness tests

To ensure the robustness of our findings, we conducted a series of additional tests (see Appendix A). First, we employed an alternative measure of AI innovation. Following established practice (Trajtenberg, 1990; Hall et al., 2005), patent citations provide a refined proxy for innovation quality, capturing technological relevance and market visibility. Accordingly, we measured the cumulative number of forward citations received by AI patents in the three years, excluding self-citations to reflect acknowledgment by the broader technological community. The

results remain consistent with our main analysis , suggesting that the effects of CEO succession extend beyond innovation quantity to encompass innovation quality.

Second, we implemented a propensity score matching (PSM) procedure to address potential sample self-selection bias arising from non-random CEO succession events (Zhang & Qu, 2016; Langan & Deuschel, 2024). Firms experiencing CEO succession were matched with firms without succession events based on identical two-digit industry codes, similar firm sizes, and comparable ROA, yielding a matched sample of 3,552 firm-year observations. The results confirm the robustness of our theoretical propositions, even after addressing potential endogeneity through matching techniques.

Third, we conducted two placebo tests by randomly assigning pseudo-treatment indicators to firm-year observations and re-estimating our models 1,000 times. The pseudo-coefficients achieved statistical significance in fewer than 5% of replications, indicating our results are not due to random chance or unobserved data structures.

Forth, we performed an Impact Threshold for a Confounding Variable (ITCV) analysis to evaluate potential omitted variable bias. The analysis reveals that an unobserved confounder would need to exhibit correlations of at least 0.228 with both the independent and dependent variables to invalidate the effect reported in Hypothesis 1a, and correlations of at least 0.180 to invalidate Hypothesis 1b. These thresholds exceed the correlations observed for all control variables included in our models, suggesting that omitted variable bias is unlikely to account for our findings.

Fifth, we conducted a reverse causality test by examining whether pre-succession AI innovation trends predict the likelihood of appointing academic successors. Specifically, we defined the independent variable as the growth rate of AI innovation in the three years preceding the succession event (measured from year $t-3$ to $t-1$) and estimated a Probit model in which the

dependent variable is the occurrence of an NACA-to-ACA CEO succession. The estimated coefficients are statistically insignificant, indicating that pre-succession growth in AI innovation does not influence the likelihood of appointing an academic CEO.

Finally, we extended our analysis to the full sample, including both high-tech and non-high-tech firms. The results confirm that the observed effects of NACA-to-ACA CEO succession on AI innovation are not confined to high-tech industries, suggesting the broader generalizability of our findings across different competitive contexts. Collectively, these tests provide strong support for the robustness and validity of our conclusions.

4.3 Further analysis

We conducted several further analyses to explore the mechanisms underlying our hypothesized relationships (see Table 3). First, our argumentation posits that academic CEOs function as cognitive catalysts, promoting AI strategy through knowledge-intensive leadership. This effect tends to be strengthened if CEOs come to the position with more external knowledge. Accordingly, we examined whether academic CEOs had prior board experience in firms led by academic peers—either during their current tenure or immediately before succession. Subsample analysis reveals that NACA-to-ACA succession significantly enhances AI innovation when CEOs with peer firm board experience ($\beta = 0.554$, $p < 0.01$), suggesting that external knowledge embedded in inter-firm ties strengthens the cognitive advantages of academic leadership.

Second, we assessed the link between CEO succession and firm AI orientation, which reflects the strategic commitment to AI, emphasizing the growth potential and comprehensive deployment of AI innovation projects (Pinski et al., 2024). Results indicate that NACA-to-ACA CEO successions are positively associated with firm AI orientation ($\beta = 2.798$, $p < 0.1$), while ACA-to-NACA transitions exhibit a negative relationship ($\beta = -1.827$, $p < 0.01$). These findings suggest

that academic CEOs facilitate strategic consensus and shared cognition on AI development, thereby enhancing the firm's prioritization of AI innovation.

Third, we examined the confounding effects of potential factors on AI innovation. Firms led by academic CEOs may be university-affiliated, and thus possess inherent advantages in technology commercialization (Ju et al., 2023). To isolate the effects of academic leadership from ownership structure, we controlled for firms whose majority shareholder is a university. The positive effect of NACA-to-ACA CEO successions on AI innovation remains significant, indicating that academic leadership, rather than university shareholder, is the key driver. Furthermore, academic CEOs may possess R&D experience, which could stimulate AI innovation and bias our interpretation. Following Jiang et al. (2025), we re-estimated the model after excluding CEOs with prior R&D experience. The results remain unchanged, confirming that academic experience drives the observed innovation-enhancing effect..

Fourth, we examined the heterogeneous effects associated with the depth and quality of CEOs' academic experience (Ju et al., 2023), specifically focusing on: (1) Academic specialization in natural sciences, which provides domain-specific technical knowledge directly applicable to AI development; (2) Academic prestige, measured as prior employment at elite universities (985/211) or national-level research institutions and academic associations, reflecting access to high-quality academic networks and research resources; and (3) Academic title, indicated by holding senior or associate senior professional titles, signaling recognized expertise and scholarly authority. The results show that all three variables significantly strengthen the positive relationship between NACA-to-ACA CEO succession and AI innovation ($\beta = 0.659$, $p < 0.01$; $\beta = 0.664$, $p < 0.01$; $\beta = 0.528$, $p < 0.05$). Collectively, these findings further confirm that high-level academic experience enhances CEOs' ability to leverage scientific knowledge and guide AI-related strategic decisions.

[Insert Table 3 about here.]

5. Discussion

5.1 Conclusion

Drawing on the knowledge-based view, we investigate how CEO succession involving changes in academic experience influences post-succession AI innovation. Our findings reveal that NACA-to-ACA CEO succession significantly enhance AI innovation, while ACA-to-NACA CEO succession attenuate it. These effects are contingent upon both internal and external knowledge configurations. TMT power with digital expertise strengthens the positive effects of appointing academic CEO and mitigates the negative consequences associated with their departure. Similarly a well-developed regional AI knowledge ecosystem moderates these relationships by providing complementary external knowledge resources and network linkages.

5.2 Theoretical contributions

This study makes several important theoretical contributions. Firstly, we advance the KBV by identifying CEO academic experience as a micro-foundational driver of firm-level knowledge orchestration. Traditional KBV research emphasizes collective routines and organizational knowledge as key sources of competitive advantage (Grant, 1996; Nonaka, 1994), yet largely overlooks the role of individual leaders in activating and reconfiguring firm-level knowledge assets. Responding to recent calls for incorporating behavioral microfoundations into knowledge-based theorizing (Ko & Choi, 2023; Nag & Gioia, 2012), we conceptualize academic CEOs as "cognitive catalysts" who orchestrate knowledge absorption, recombination, and governance through distinct epistemic orientations. By bridging individual leadership traits with organizational knowledge processes, our integrative framework enriches the behavioral micro-foundations of

KBV.

Secondly, we contribute to the emerging literature on AI innovation by positioning CEO academic experience as a critical endogenous driver. Prior studies have predominantly conceptualized AI as an external technological enabler or emphasized firm-level technological capabilities (Mariani et al., 2023), while the internal human-centric mechanisms that facilitate AI innovation remain underexplored. We demonstrate that academic CEOs are positively associated with AI patent output and impact, highlighting the cognitive foundations of innovation in the AI domain. Our research advances the discourse on AI innovation antecedents (Igna & Venturini, 2023) and addresses a critical gap in understanding how leadership cognition drives knowledge-intensive innovation.

Thirdly, we extend the CEO succession literature by reframing leadership transition as a strategic inflection point that reshapes firms' innovation trajectories. By introducing the concept of the "experience change effect" and jointly considering the academic experience of both predecessors and successors, we adopt a dynamic perspective to examine how leadership transitions systematically reshape strategic decisions and innovation outcomes. Previous research has primarily emphasized the disruptive consequences of CEO succession, such as performance decline and organizational instability (Wu et al., 2025; Zhang & Qu, 2016). Our findings reveal that NACA-to-ACA CEO successions promote AI innovation, enriching existing research on the dynamic effects of leadership configuration in knowledge-intensive innovation.

Finally, we develop a multilevel contingency framework to delineate the boundary conditions under which CEO academic experience drives AI innovation. While prior research has examined isolated moderators of CEO succession effects (Wu et al., 2025), few studies have integrated both internal and external contingencies within a unified framework. We demonstrate that TMT power

with digital expertise not only strengthens the innovation-enhancing effects of academic CEOs but also attenuates potential declines during ACA-to-NACA transitions. Externally, a robust regional AI knowledge ecosystem serves as both a buffer and amplifier, providing knowledge spillovers and network linkages that sustain innovation through leadership change (De Noni et al., 2018). Collectively, this integrative framework contributes to behavioral strategy and contingency theory by clarifying the applicable scope under which CEO cognitive traits promote tangible innovation outcomes.

5.3 Practical implications

This study provides actionable insights for boards and managers involved in CEO succession planning, particularly in knowledge-intensive sectors such as AI. Boards are advised to critically assess the firm's strategic needs during leadership transitions, paying particular attention to whether incoming CEO candidates possess academic experience. Appointing academic CEOs can represent a forward-looking investment in the firm's knowledge-generation capabilities and long-term technological advancement, particularly for firms seeking to promote AI innovation. However, boards should recognize that the effectiveness of academic CEOs depends critically on supportive internal and external conditions. Specifically, boards should assess whether the firm's TMT possesses sufficient digital expertise to complement and implement the academic CEO's innovation vision. When such expertise is lacking, boards may consider restructuring the TMT composition prior to or concurrent with the succession event, recruiting executives with digital backgrounds to create a more conducive environment for AI innovation. Additionally, boards should evaluate the firm's regional context, as academic CEOs may achieve greater impact in robust AI knowledge ecosystems that provide research talent, collaborative networks, and knowledge spillovers. When transitioning from an academic to a non-academic CEO, boards

should implement deliberate continuity mechanisms to safeguard innovation momentum. These mechanisms might include establishing formal advisory roles for departing academic CEOs to preserve institutional knowledge, forming strategic alliances with AI-focused firms to maintain external knowledge flows, or strengthening partnerships with universities and research institutions to compensate for the reduced internal academic orientation.

For newly appointed CEOs, our findings offer guidance on leveraging personal experience and organizational resources to drive AI innovation. Academic CEOs should recognize that their innovative cognition, scientific knowledge, and research networks constitute valuable strategic assets capable of shaping the firm's innovation trajectory. To maximize these advantages, academic CEOs should actively engage TMT members with digital expertise, fostering synergies between scientific insights and technological implementation. They can further leverage institutional affiliations by initiating joint research projects with universities, recruiting talent through academic networks, and cultivatin a culture that values exploratory learning and long-term knowledge creation. Meanwhile, academic CEOs should remain attentive to potential tensions between scientific pursuits and commercial pressures, balancing exploratory innovation with operational imperatives that sustain firm performance. For non-academic CEOs succeeding academic predecessors, our findings highlight the importance of proactively addressing potential knowledge gaps. These CEOs should prioritize retaining and empowering TMT members with digital expertise, who can serve as critical bridges for maintaining the firm's AI innovation capabilities. Non-academic CEOs may also benefit from establishing external knowledge channels, such as advisory boards comprising academic experts, strategic partnerships with research institutions, or active engagement in regional AI ecosystems. By deliberately cultivating these internal and external knowledge resources, non-academic CEOs can sustain innovation momentum

despite lacking personal academic experience.

5.4 Limitations and future research

Despite its contributions, this study has several limitations that offer fruitful directions for future research. First, our reliance on firm-level data from a single-country context (China) may limit the generalizability of the findings. China presents a unique institutional environment characterized by strong government support for AI development, close university-industry linkages, and a cultural emphasis on academic credentials in leadership selection. These features may amplify the influence of CEO academic experience on AI innovation relative to developed economies, where market-driven mechanisms and different academic-industry boundaries prevail. Future studies should examine cross-national samples, especially comparing developed and developing economies or culturally distinct regions. Such comparative research could provide deeper insights into how institutional and cultural contexts mediate the influence of CEO academic experience on innovation outcomes.

Second, although our model accounts for a comprehensive set of organizational and contextual controls, other important contingencies likely affect the relationship between CEO succession and AI innovation. For instance, CEO–board relationships may influence the latitude academic CEOs have to implement innovation agendas; and firms at different life-cycle stages may exhibit varying receptivity to the exploratory orientation that academic CEOs bring. Additionally, the role of university-industry collaborations merit closer examination, as academic CEOs may leverage their institutional networks differently depending on prior affiliations and ongoing research partnerships. Future research can investigate how these factors interact with CEO academic experience to amplify or dampen its influence on AI innovation.

Third, our reliance on archival data limits the ability to explore the motivations, cognitive

orientations, and strategic intent of CEOs during succession. Leadership transitions involve complex psychological and behavioral processes that are not readily captured by quantitative indicators. For instance, we cannot capture how academic CEOs navigate tensions between scientific pursuits and commercial pressures, communicate innovation visions to potentially resistant TMT members, or adapt their leadership styles in response to implementation challenges. Future research could employ qualitative methods—including in-depth interviews, longitudinal case studies, or executive surveys—to better understand how CEOs interpret their roles and enact innovation strategies. Such mixed-method approaches would offer a richer, more nuanced understanding of leadership and innovation dynamics.

Fourth, our measurement of AI innovation through patent applications, while consistent with established practice, may not capture the full spectrum of AI-related innovative activities. Patents primarily reflect codified and protectable knowledge outputs, but may underrepresent organizational innovations in AI deployment or exploratory AI initiatives that have not generated formal intellectual property. Future research could complement patent-based measures with alternative indicators such as AI-related product releases, AI-driven revenues, or collaborative AI R&D agreements. Furthermore, examining the temporal dynamics of CEO influence on AI innovation would enrich our findings, including potential lagged effects beyond our three-year window and the persistence of innovation trajectories.

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Table 1
Descriptive statistics and correlations matrix.

Variable	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. AI innovation	2.124	1.786	1															
2. NACA-to-ACA CEO succession	0.068	0.252	0.070*	1														
3. ACA-to-NACA CEO succession	0.172	0.377	-0.090*	-0.123*	1													
4. TMT power with digital expertise	0.366	0.286	0.300*	0.021	0.017	1												
5. Regional AI knowledge ecosystem	7.374	1.873	0.343*	0.101*	0.019	0.173*	1											
6. Predecessor tenure	1.502	0.557	0.004	-0.040	0.061*	-0.004	-0.048*	1										
7. Successor duality	0.145	0.352	-0.046	0.250*	-0.065*	-0.060*	0.007	0.007	1									
8. Successor gender	0.941	0.236	0.067*	0.001	0.007	0.053*	-0.037	-0.017	-0.039	1								
9. Board independence	0.375	0.056	0.064*	0.038	0.007	0.041	0.063*	-0.019	0.063*	-0.105*	1							
10. TMT size	2.370	0.249	0.196*	-0.007	-0.013	0.018	-0.012	-0.027	-0.161*	0.125*	-0.292*	1						
11. Share of graduated employees	0.307	0.202	0.391*	0.076*	-0.020	0.326*	0.390*	-0.038	0.011	0.012	0.090*	0.054*	1					
12. Ownership concentration	0.323	0.142	0.047*	0.015	0.043	-0.025	0.004	-0.029	-0.057	0.026	0.052*	0.024	-0.020	1				
13. Firm size	22.210	1.240	0.457*	-0.059*	-0.062*	0.007	0.068*	0.038	-0.122*	0.067*	-0.007	0.355*	0.033	0.191*	1			
14. Pre-succession firm performance	0.033	0.061	0.058*	0.013	0.022	-0.003	0.007	0.042	-0.008	0.001	-0.039	0.032	0.064*	0.150*	-0.003	1		
15. R&D intensity	0.047	0.039	0.284*	0.035	-0.020	0.253*	0.213*	0.016	0.041	-0.025	0.062*	-0.055*	0.395*	-0.159*	-0.254*	0.035	1	
16. Market competition	0.129	0.135	0.002	-0.024	0.083*	-0.002	0.131*	-0.040	-0.048*	0.043	-0.027	0.039	0.048*	0.001	0.086*	0.013	-0.122*	1

Note: N = 1,776; Absolute correlation coefficients exceeding 0.047 are significant at the 5% level ($p < 0.05$).

Table 2
OLS Estimates on post-succession AI innovation.

Variables	Model 1	Model 2	Model 4	Model 5	Model 6
NACA-to-ACA CEO succession		0.451***	0.332***	0.304***	0.245**
		(0.111)	(0.104)	(0.103)	(0.097)
ACA-to-NACA CEO succession		-0.279***	-0.293***	-0.268***	-0.284***
		(0.086)	(0.084)	(0.084)	(0.082)
NACA-to-ACA CEO succession × TMT power with digital expertise			1.859***		1.793***
			(0.302)		(0.323)
ACA-to-NACA CEO succession × TMT power with digital expertise			0.534**		0.545**
			(0.256)		(0.250)
TMT power with digital expertise			0.644***		0.631***
			(0.132)		(0.129)
NACA-to-ACA CEO succession × Regional AI knowledge ecosystem				0.111**	0.039
				(0.049)	(0.048)
ACA-to-NACA CEO succession × Regional AI knowledge ecosystem				0.102***	0.095***
				(0.037)	(0.037)
Regional AI knowledge ecosystem				0.164***	0.161***
				(0.023)	(0.023)
Predecessor tenure	-0.134	-0.111	-0.141	-0.065	-0.097
	(0.123)	(0.122)	(0.119)	(0.119)	(0.116)
Successor duality	-0.193*	-0.356***	-0.273**	-0.253**	-0.177
	(0.114)	(0.122)	(0.119)	(0.119)	(0.116)
Successor gender	0.183	0.162	0.149	0.219*	0.205*
	(0.127)	(0.128)	(0.127)	(0.125)	(0.124)
Board independence	1.356**	1.260**	1.210*	1.356**	1.324**
	(0.642)	(0.630)	(0.623)	(0.607)	(0.597)
TMT size	0.233	0.175	0.179	0.237	0.246
	(0.169)	(0.168)	(0.166)	(0.163)	(0.161)
Share of graduated employees	2.105***	1.957***	1.872***	1.682***	1.612***
	(0.292)	(0.296)	(0.292)	(0.288)	(0.287)
Ownership concentration	-0.100	-0.103	-0.089	-0.154	-0.140
	(0.283)	(0.282)	(0.277)	(0.279)	(0.275)
Firm size	0.777***	0.785***	0.767***	0.769***	0.750***
	(0.039)	(0.039)	(0.039)	(0.038)	(0.038)
Pre-succession firm performance	1.610***	1.663***	1.578***	1.571***	1.500***
	(0.583)	(0.578)	(0.572)	(0.553)	(0.545)
R&D intensity	9.470***	9.362***	9.168***	8.770***	8.568***
	(1.117)	(1.097)	(1.051)	(1.078)	(1.035)
Market competition	-0.893**	-0.822*	-0.754*	-0.988**	-0.909**
	(0.448)	(0.442)	(0.392)	(0.424)	(0.379)
Inverse Mill's ratio on experiencing CEO succession	0.267	0.246	0.328	0.168	0.257
	(0.258)	(0.256)	(0.252)	(0.250)	(0.246)
Inverse Mill's ratio on choosing a CEO with academic experience	-0.296***	-0.385***	-0.306***	-0.266***	-0.194*
	(0.091)	(0.101)	(0.100)	(0.100)	(0.099)

Constant	-17.212***	-16.991***	-16.612***	-16.849***	-16.477***
	(0.859)	(0.848)	(0.843)	(0.833)	(0.827)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Adj.R-squared	0.499	0.506	0.518	0.527	0.538

Note: N = 1,776; Standardized coefficients are reported with clustered robust standard errors in parentheses.
* p < .1, ** p < 0.05, *** p < 0.01

Table 3
Further analysis.

Variables	Model 1 CEOs with board experience at academic peer firms	Model 2 CEOs without board experience at academic peer firms	Model 3 DV: Firm AI orientation	Model 4 Additional control: University ownership	Model 5 Excluding CEO with R&D experience	Model 6 Moderator: Academic specialization	Model 7 Moderator: Academic prestige	Model 8 Moderator: Academic title
NACA-to-ACA CEO succession	0.554***	0.511**	2.798*	0.447***	0.331**	0.240*	0.251**	0.297**
	(0.160)	(0.243)	(1.446)	(0.111)	(0.138)	(0.129)	(0.126)	(0.125)
ACA-to-NACA CEO succession	-0.107	-0.153**	-1.827***	-0.273***	-0.189***	-0.277***	-0.277***	-0.279***
	(0.154)	(0.075)	(0.611)	(0.085)	(0.070)	(0.086)	(0.086)	(0.086)
University shareholder				0.245				
				(0.370)				
NACA-to-ACA CEO succession × Research experience						0.659***		
						(0.233)		
NACA-to-ACA CEO succession × Institutional prestige							0.664***	
							(0.238)	
NACA-to-ACA CEO succession × Academic title								0.528**
								(0.245)
Predecessor tenure	-0.410	-0.023	-0.294	-0.113	0.097	-0.115	-0.109	-0.120
	(0.289)	(0.105)	(1.213)	(0.123)	(0.102)	(0.123)	(0.123)	(0.123)
Successor duality	-0.520**	-0.298**	-0.078	-0.342***	-0.207*	-0.330***	-0.338***	-0.347***
	(0.219)	(0.124)	(1.295)	(0.123)	(0.121)	(0.123)	(0.123)	(0.122)
Successor gender	0.193	0.031	0.652	0.165	0.001	0.159	0.160	0.160
	(0.231)	(0.104)	(1.056)	(0.127)	(0.094)	(0.127)	(0.127)	(0.127)
Board independence	2.122	0.267	8.420	1.259**	0.610	1.257**	1.284**	1.277**
	(1.438)	(0.642)	(5.578)	(0.630)	(0.591)	(0.626)	(0.626)	(0.629)
TMT size	0.839**	0.324*	0.324	0.163	0.414**	0.154	0.154	0.162
	(0.351)	(0.172)	(1.381)	(0.168)	(0.163)	(0.168)	(0.168)	(0.168)
Share of graduated	1.626***	1.177***	5.632**	1.951***	1.087***	1.947***	1.950***	1.955***

employees	(0.608)	(0.293)	(2.228)	(0.296)	(0.266)	(0.295)	(0.295)	(0.296)
Ownership concentration	-0.175	0.242	-5.810**	-0.100	0.324	-0.100	-0.097	-0.108
	(0.525)	(0.298)	(2.364)	(0.283)	(0.273)	(0.282)	(0.283)	(0.283)
Firm size	0.329***	0.335***	0.848***	0.786***	0.326***	0.782***	0.782***	0.781***
	(0.083)	(0.041)	(0.306)	(0.039)	(0.040)	(0.039)	(0.039)	(0.039)
Pre-succession firm performance	1.814	1.990***	-5.012	1.654***	2.354***	1.626***	1.632***	1.599***
	(1.112)	(0.546)	(5.083)	(0.578)	(0.494)	(0.575)	(0.575)	(0.577)
R&D intensity	7.964***	5.267***	23.062**	9.414***	5.066***	9.176***	9.181***	9.215***
	(2.449)	(1.330)	(11.011)	(1.092)	(1.263)	(1.093)	(1.093)	(1.094)
Market competition	-0.299	-0.636*	-4.842	-0.810*	-0.676*	-0.820*	-0.825*	-0.810*
	(0.736)	(0.361)	(4.919)	(0.443)	(0.364)	(0.450)	(0.451)	(0.447)
Inverse Mill's ratio on experiencing CEO succession	1.123*	0.102	0.235	0.249	-0.159	0.255	0.248	0.267
	(0.626)	(0.231)	(2.442)	(0.257)	(0.208)	(0.257)	(0.257)	(0.258)
Inverse Mill's ratio on choosing a CEO with academic experience	-0.470**	-0.274***	-2.233**	-0.370***	-0.278***	-0.375***	-0.376***	-0.378***
	(0.198)	(0.098)	(0.870)	(0.101)	(0.095)	(0.102)	(0.102)	(0.101)
Constant	-9.203***	-7.945***	-17.594**	-0.113	-8.124***	-16.857***	-16.867***	-16.847***
	(2.090)	(0.911)	(7.240)	(0.123)	(0.882)	(0.847)	(0.848)	(0.854)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	359	1417	1776	1776	1641	1776	1776	1776
Adj.R-squared	0.398	0.281	0.270	0.506	0.281	0.508	0.508	0.507

Note: Models 1-2 present subsample analyses based on CEO board experience at academic peer firms. Model 3 examines the relationship between CEO succession involving changes in academic experience and firm AI orientation. Model 4 presents supplemental analyses for university ownership. Models 5 re-estimates the models after excluding CEO successions involving CEOs with prior R&D experience. Models 6-8 test the moderating effects of academic specialization, academic prestige, and academic title.

N = 1,776; Standardized coefficients are reported with clustered robust standard errors in parentheses.

* p < .1, ** p < 0.05, *** p < 0.01

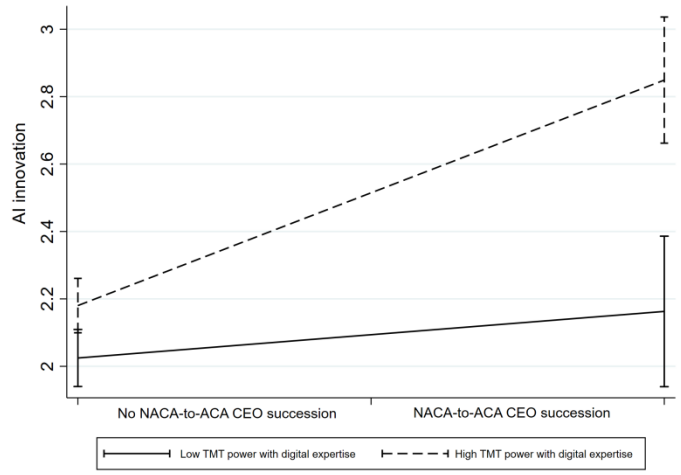

Fig. 1. Moderating effect of TMT power with digital expertise (H2a).

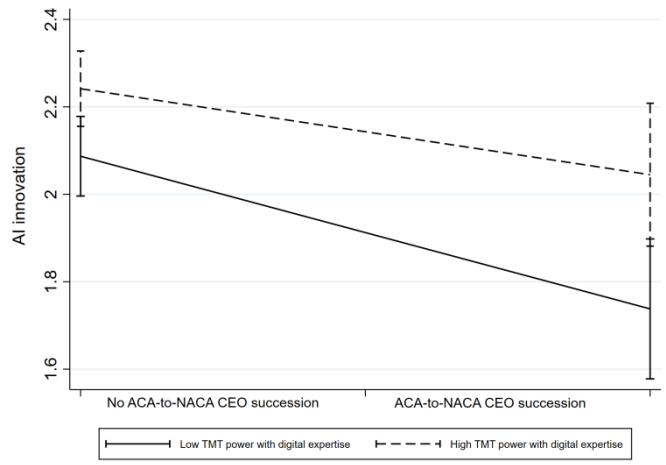

Fig. 2. Moderating effect of TMT power with digital expertise (H2b).

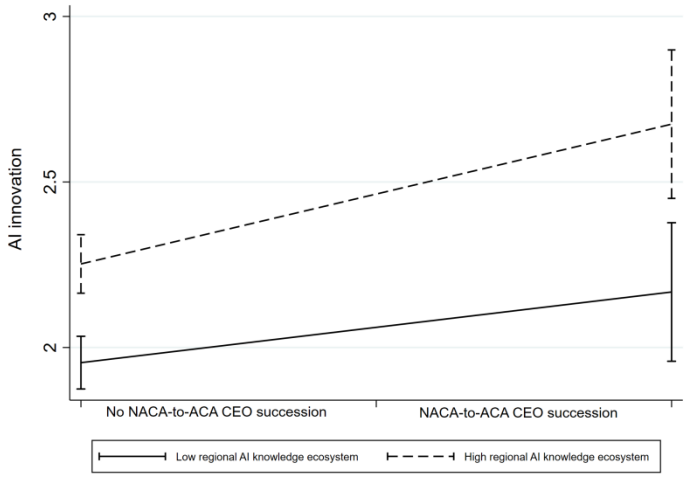

Fig. 3. Moderating effect of regional AI knowledge ecosystem (H3a).

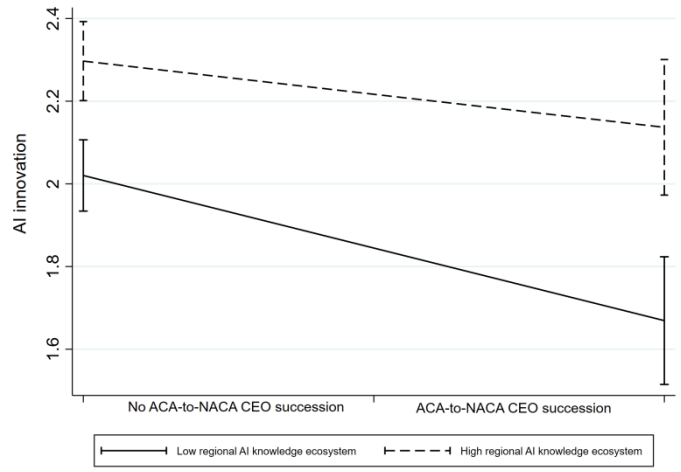

Fig. 4. Moderating effect of regional AI knowledge ecosystem (H3b).

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Supplementary material
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